



EE 354/CE 361/MATH 310 -
Introduction to Probability
and Statistics
Fall 2025



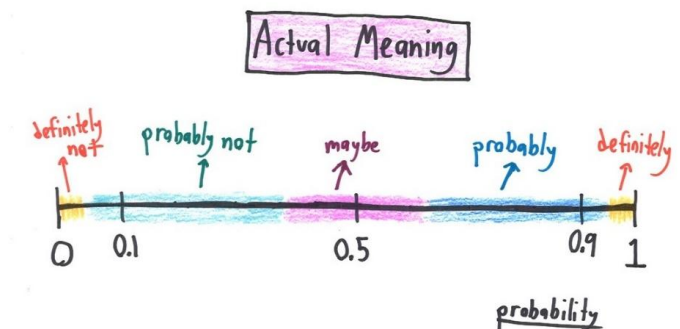


EE 354/CE 361/MATH 310 L2 section (Fall 2025)

Introduction to Probability and Statistics -

Aamir Hasan

Week 8 - Lecture No 14 – October 06, 2025



Announcements!

1. Lectures up to No 13 posted. All quiz scores posted.
2. To enable N-2. we will like to have a quiz No 4 in the next class (DRV)
3. Midterm Exam - October 8, 2025, from 05:40 PM to 06:55 PM – **venue H-507 (L2)** – Good Luck
 - **Instructions:**
 1. Mobile phones and other electronic devices must be turned off and put away before the start of the exam. You are allowed to use a scientific calculator (Not the one on your smart phone. Sharing of calculators is not allowed as well).
 2. No food or drink, other than water, is allowed in the examination room.
 3. You are not allowed to communicate with other candidates.
 4. You are not allowed to leave your seat unless granted permission by an invigilator.
 5. Assume where necessary and justify your assumptions.
 6. You are required to fill all answers in the provided answer booklet.
 7. You are allowed to bring a cheat sheet. It must be a handwritten A4 sheet of paper. You may write on both sides of the sheet. Photocopied cheat sheets are not allowed.

Agenda for today

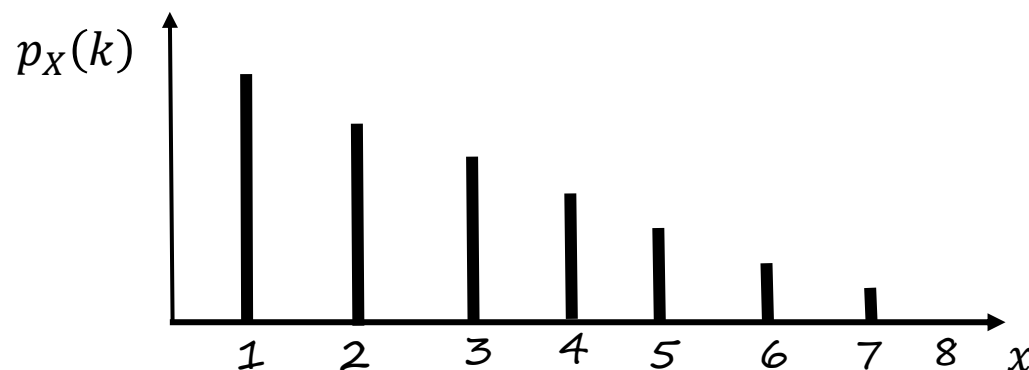
1. Unit 4 - Discrete Random Variables

Unit 4
Discrete Random Variables

Conditioning a geometric R.V.

- **X**: number independent tosses of a coin until first heads with $P(\text{Heads}) = p$

$$p_X(x) = (1 - p)^{x-1}p \quad x = 1, 2, 3, \dots$$



Memorylessness:

Number of **remaining** coin tosses, conditioned on Tails in the first toss, is **Geometric**, with parameter p

Conditioned on $X > 1$, $X - 1$ is geometric with parameter p

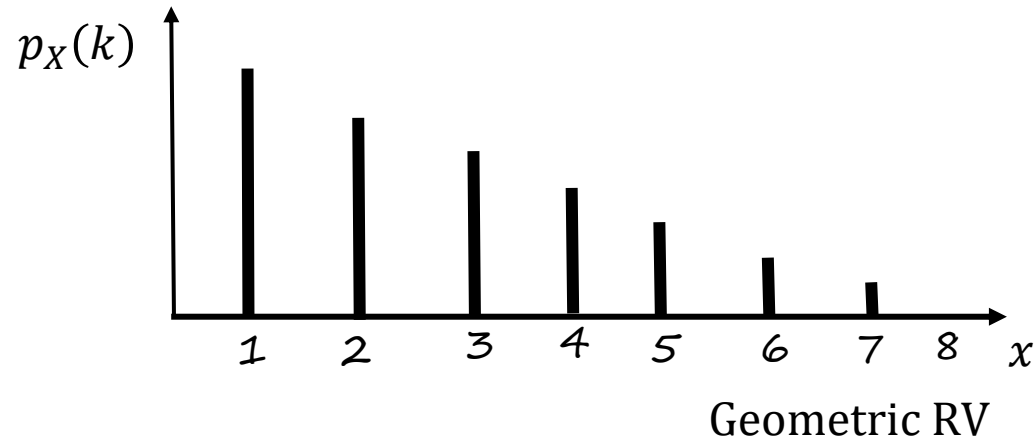
$$\begin{aligned} P_{X-1|X>1}(3) &= P(X - 1 = 3 \mid X > 1) = P(T_2 T_3 H_4 \mid T_1) = P(T_2 T_3 H_4) \\ &= p(1-p)^2 \\ &= P_X(3) \end{aligned}$$

$$P_{X-1|X>1}(k) = P_X(k)$$

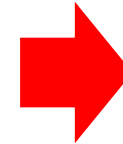
$$P_{X-n|X>n}(k) = P_X(k)$$

Conditioned on $X > n$, $X - n$ is geometric with parameter p

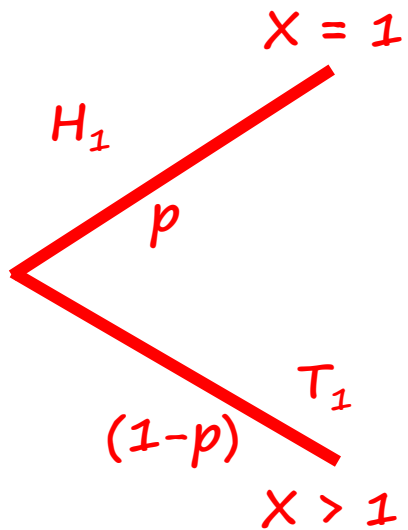
The mean of the geometric



$$E[X] = \sum_{k=1}^{\infty} k p_X(k) = \sum_{k=0}^n k(1-p)^{k-1}p$$



$$E[X] = \frac{1}{p}$$



$$\Rightarrow E[X] = 1 + E[X - 1]$$

$$\Rightarrow = 1 + p E[X - 1 \mid X = 1] + (1-p) E[X - 1 \mid X > 1]$$

$$\Rightarrow E[X] = 1 + 0 + (1-p) E[X]$$

$$\Rightarrow E[X] = 1 + 0 + E[X] - p \cdot E[X]$$

Exercise



Equally
likely

Problem 3 (25 points): A Random Variable Z_n is defined as $Z_n = \max(X_1, X_2, \dots, X_n)$, where X_i 's are outcomes of identical and independently distributed tetrahedral die (4-sided). As an example $Z_2 = \max(X_1, X_2)$ - meaning Z_2 is the maximum of the outcome of two independent throws of tetrahedral die X_1 and X_2 .

Part a (5 points): Calculate probability $P(Z_{10} < 4)$.

Part b (5 points): Calculate probability $P(Z_3 = 2)$.

Part c (5 points): Calculate and plot the probability mass function, PMF of Z_2 .

Part d (10 points): Find the Expected value, $E[Z_2|A]$, where event $A = \{X_1 = 3\}$.

Part a

$$Z_{10} < 4 \Rightarrow \max(X_1, \dots, X_{10}) < 4$$

means 0 zero No' of 4 in 10 throws

$$\begin{aligned} \text{Probability of No 4 in one throw} &= 1 - \frac{1}{4} \\ &= \frac{3}{4} \end{aligned}$$

$$\text{So } P(Z_{10} < 4) = \left(\frac{3}{4}\right)^{10} = 0.0563$$

Exercise



Equally likely

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Part b

$$P(Z_3 = 2)$$

$$= \max(X_1, X_2, X_3) = 2$$

$\Downarrow$

for $Z_3 = 2$

outcomes
1, 1, 2
1, 2, 1
2, 1, 1
2, 2, 1
2, 1, 2
1, 2, 2
2, 2, 2

Each Seq. probability =

1	2	2	✓
2	1	2	
2	2	1	

$$= 7 \times \frac{1}{4^3}$$

$$= 0.109$$

Exercise



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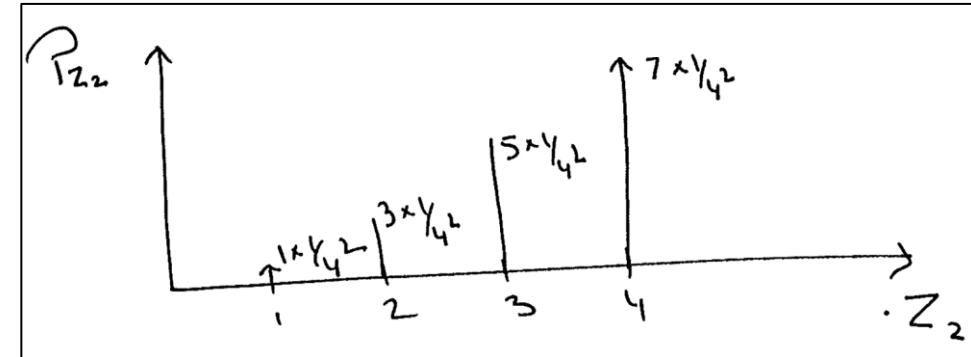
Part d (10 points): Find the Expected value, $E[Z_2|A]$, where event $A = \{X_1 = 3\}$.

Part c

$Z_2 = 1 \Rightarrow$	$X_1 \quad X_2$	
	1 1	$1 \times \frac{1}{4^2}$
$Z_2 = 2 \Rightarrow$	1 2	
	2 1	$3 \times \frac{1}{4^2}$
	2 2	

$Z_2 = 3 \Rightarrow$	1 3	
	2 3	
	3 3	$5 \times \frac{1}{4^2}$
	3 1	
	3 2	

$Z_2 = 4 \Rightarrow$	1 4	
	2 4	
	3 4	
	4 4	$7 \times \frac{1}{4^2}$
	4 1	
	4 2	
	4 3	



Exercise



Equally likely

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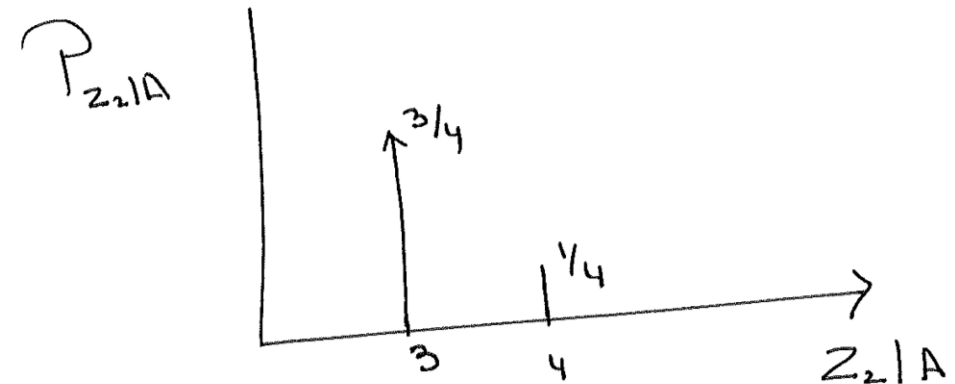
Part c (5 points): Calculate and plot the probability mass function, PMF of Z_2 .

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Part d

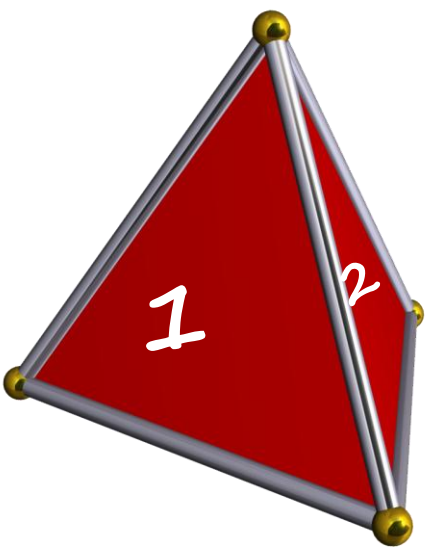
$$E[Z_2 | (X_1 = 3)]$$

X_1	X_2	$P(X_2 = x)$	Z_2
3	1	$\frac{1}{4}$	3
3	2	$\frac{1}{4}$	3
3	3	$\frac{1}{4}$	3
3	4	$\frac{1}{4}$	4



$$\begin{aligned} E[Z_2|A] &= 3 \times \frac{3}{4} + 4 \times \frac{1}{4} \\ &= \frac{9}{4} + \frac{4}{4} = \frac{13}{4} \end{aligned}$$

Exercise!



Equally
likely

A Random Variable Z_n is defined as $Z_n = \min (X_1; X_2; \dots ; X_n)$, where X_i s are outcomes of identical and independently distributed tetrahedral die (4-sided). As an example $Z_2 = \min (X_1; X_2)$ - meaning Z_2 is the minimum of the outcome of two independent throws of tetrahedral die X_1 and X_2 .

$$\text{var}(X) = E[X^2] - (E[X])^2$$

1. Calculate Variance of Z_1
2. Calculate Variance of Z_2

Multiple random variables and joint PMFs

- $X : p_X$
 - $Y : p_Y$
- marginal pmfs*
- $P(X = Y) = 2/20$

Joint PMF: $p_{X,Y}(x, y) = P(X = x \text{ and } Y = y)$

Y =
Second roll

4	1/20	2/20	2/20	
3	2/20	4/20	1/20	2/20
2		1/20	3/20	1/20
1		1/20		
	1	2	3	4

X = First roll $P_X(3)$

$$\sum_x \sum_y p_{X,Y}(x, y) = 1$$

$$P_{X,Y}(1, 3) = 2/20$$

$$P_X(4) = 1/20 + 2/20$$

$$P_Y(2) = (1 + 3 + 1)/20$$

$$p_X(x) = \sum_y p_{X,Y}(x, y)$$

$$p_Y(y) = \sum_x p_{X,Y}(x, y)$$